Fundamentals of Data Science

Assignment 1

Name: Praptee Dewan

Student ID: 25068590

Module Code: UFCFK1-15-0

Instructor: Kaushal Rimal

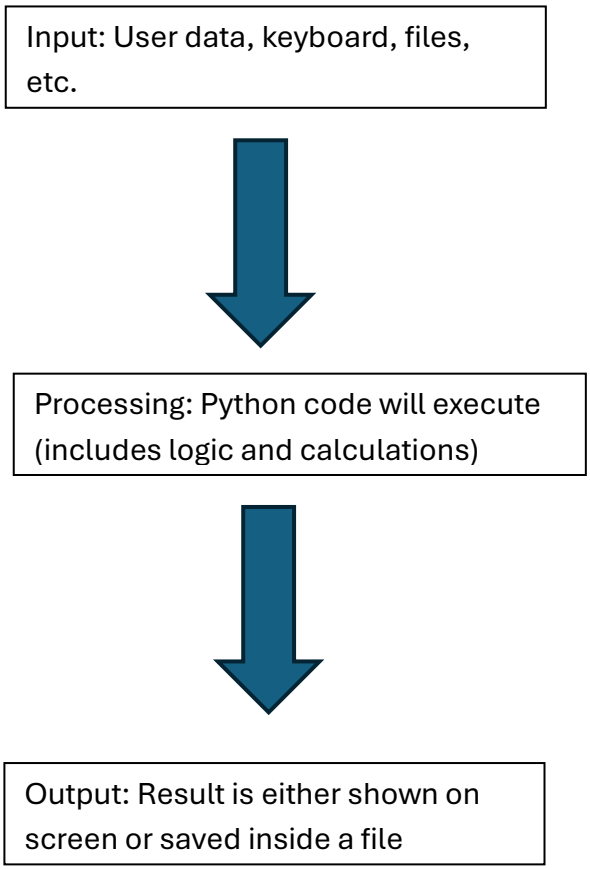
Submission Date: 17 April 2026

Github link: https://github.com/dprapti25-coder/FODS-Submissions.git

1) Flowchart of Python Working

This diagram shows how Python works using three stages: input, processing, and output.

- **Flowchart:**



2) Arithmetic Operations

This program takes two integers from the user and performs addition, multiplication, division, modulus, and exponentiation.

- Code:

```
''This program takes user input for two integers and performs basic arithmetic operations'''
a = int(input("Enter first number: "))
```

```
b = int(input("Enter second number: "))
print("Addition:", a + b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Modulus:", a % b)
print("Exponentiation:", a ** b)
```

- **Output Screenshot:**

```
PS C:\Users\praka\OneDrive\Desktop\TBC\FODS\TAKE HOME ASSESSMENT SUBMISSION> & C:\Users\praka\AppData\Local\Programs\Python
\Python314\python.exe "c:/Users/praka/OneDrive/Desktop/TBC/FODS/TAKE HOME ASSESSMENT SUBMISSION/Assignment1/numops.py"
Enter first number: 1
Enter second number: 2
Addition: 3
Multiplication: 2
Division: 0.5
Modulus: 1
Exponentiation: 1
PS C:\Users\praka\OneDrive\Desktop\TBC\FODS\TAKE HOME ASSESSMENT SUBMISSION>
```

3) Number Classification

This program checks whether a number is positive/negative and even/odd.

- **Code:**

```
'''Classifies numbers as positive negative even or odd'''
num = int(input("Enter a number: "))

if num > 0:
    if num % 2 == 0:
        print("Positive Even")
    else:
        print("Positive Odd")
elif num < 0:
    if num % 2 == 0:
        print("Negative Even")
    else:
        print("Negative Odd")
else:
    print("Zero")
```

- **Output Screenshot:**

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\praka\OneDrive\Desktop\TBC\FODS\TAKE HOME ASSESSMENT SUBMISSION> & C:\Users\praka\AppDat
a\Local\Programs\Python\Python314\python.exe "c:/Users/praka/OneDrive/Desktop/TBC/FODS/TAKE HOME ASS
ESSMENT SUBMISSION/Assignment1/numclass.py"
Enter a number: 3
Positive Odd
PS C:\Users\praka\OneDrive\Desktop\TBC\FODS\TAKE HOME ASSESSMENT SUBMISSION>
```

4) Mathematical Operations

This program calculates cube, cube root, logarithms, and power of a number.

- **Code:**

```python
import math

num = float(input("Enter a number: "))

print("Cube:", num**3)
print("Cube root:", num**(1/3))
print("Natural log:", math.log(num))
print("Log base 2:", math.log2(num))
print("Power 6:", num**6)
```

- **Output Screenshot:**

```
PS C:\Users\praka\OneDrive\Desktop\TBC\FODS\TAKE HOME ASSESSMENT SUBMISSION> & C:\Users\praka\AppDat
a\Local\Programs\Python\Python314\python.exe "c:/Users/praka/OneDrive/Desktop/TBC/FODS/TAKE HOME ASS
ESSMENT SUBMISSION/Assignment1/mathops.py"
Enter a number: 5
Cube: 125.0
Cube root: 1.7099759466766968
Natural log: 1.6094379124341003
Log base 2: 2.321928094887362
Power 6: 15625.0
PS C:\Users\praka\OneDrive\Desktop\TBC\FODS\TAKE HOME ASSESSMENT SUBMISSION>
```

5) Marks and Grade

This program calculates total, average, percentage, and assigns a grade based on rules.

- **Code:**

```python
marks = []
for i in range(6):
    m = float(input(f"Enter marks for subject {i+1}: "))
    marks.append(m)

total = sum(marks)
avg = total / 6
percentage = avg

if percentage >= 85:
    grade = "Distinction"
elif percentage >= 70:
    grade = "First Division"
elif percentage >= 55:
    grade = "Second Division"
elif percentage >= 45:
    grade = "Third Division"
else:
    grade = "Fail"

print("Total:", total)
print("Average:", avg)
print("Percentage:", percentage)
print("Grade:", grade)
```

- **Output Screenshot:**

```
PS C:\Users\praka\OneDrive\Desktop\TBC\FODS\TAKE HOME ASSESSMENT SUBMISSION> & C:\
a\Local\Programs\Python\Python314\python.exe "c:/Users/praka/OneDrive/Desktop/TBC/
ESSMENT SUBMISSION/Assignment1/marksgrades.py"
Enter marks for subject 1: 32.33
Enter marks for subject 2: 45
Enter marks for subject 3: 69
Enter marks for subject 4: 90
Enter marks for subject 5: 76
Enter marks for subject 6: 55
Total: 367.33
Average: 61.221666666666664
Percentage: 61.221666666666664
Grade: Second Division
PS C:\Users\praka\OneDrive\Desktop\TBC\FODS\TAKE HOME ASSESSMENT SUBMISSION>
```

6) Prime Numbers in Range

This program displays all prime numbers within a given range and calculates their count and sum.

- **Code:**

```
start = int(input("Enter start: "))
end = int(input("Enter end: "))

count = 0
total = 0

for num in range(start, end+1):
    if num > 1:
        for i in range(2, num):
            if num % i == 0:
                break
        else:
            print(num)
            count += 1
            total += num

print("Count:", count)
print("Sum:", total)
```

- **Output Screenshot:**

```
PS C:\Users\praka\OneDrive\Desktop\TBC\FODS\TAKE HOME ASSESSMENT SUBMISSI
a\Local\Programs\Python\Python314\python.exe "c:/Users/praka/OneDrive/Des
ESSMENT SUBMISSION/Assignment1/primenums.py"
Enter start: 1
Enter end: 11
2
3
5
7
11
Count: 5
Sum: 28
```

7) Divisible Numbers

This program finds numbers divisible by 9 but not by 6 within a range.

- **Code:**

```
start = int(input("Enter start: "))
end = int(input("Enter end: "))

```

```
for num in range(start, end+1):
    if num % 9 == 0 and num % 6 != 0:
        print(num)
```

- **Output Screenshot:**

```
Enter start: 1
Enter end: 100
9
27
45
63
81
99
```

8) Factorial

This program calculates factorial and validates user input.

- **Code:**

```
num = input("Enter a number: ")

if num.isdigit():
    num = int(num)
    fact = 1
    for i in range(1, num+1):
        fact *= i
    print("Factorial:", fact)
else:
    print("Invalid input!")
```

- **Output Screenshot:**

```
Enter a number: 10
Factorial: 3628800
```

9) Sum of Positive and Negative Numbers

This program calculates separate sums using a loop until user exits.

- **Code:**

```
pos_sum = 0
neg_sum = 0

while True:
    num = int(input("Enter number: "))

    if num > 0:
        pos_sum += num
    else:
        neg_sum += num

    choice = input("Continue? (y/n): ")
    if choice.lower() == 'n':
        break

print("Positive sum:", pos_sum)
print("Negative sum:", neg_sum)
```

- **Output Screenshot:**

```
Enter number: 90
Continue? (y/n): y
Enter number: 6
Continue? (y/n): n
Positive sum: 96
Negative sum: 0
```

10) Number Guessing Game

This program generates a random number and allows the user to guess within limited attempts.

- **Code:**

```python
import random

number = random.randint(1, 50)

for i in range(7):
    guess = int(input("Guess number: "))

    if guess > number:
        print("Too high")
    elif guess < number:
        print("Too low")
    else:
        print("Correct!")
        break
else:
    print("Better luck next time!")
```

- **Output Screenshot:**

```
Guess number: 78
Too high
Guess number: 60
Too high
Guess number: 45
Too high
Guess number: 16
Too low
Guess number: 35
Correct!
```